

THE (ALMOST) IMPOSSIBLE PUZZLE

ABSTRACT. Tom, Jerry and Spike play the following game. Tom generates a random binary code of length n , which he sends to Spike along with a *secret number* between 0 and $n - 1$. Spike flips exactly one bit in the code and then sends only the modified binary code to Jerry. Is there a pre-established strategy for Spike and Jerry such that Jerry can determine the *secret number* solely by looking at the code sent by Spike?

1. DEFINITIONS

Definition 1.1. $\text{Fin } n := \{0, \dots, n - 1\}$, $n \in \mathbb{N}$ ($\text{Fin } 0 := \emptyset$). $\mathbb{Z}_2 := \{0, 1\}$ is the [commutative ring of integers modulo 2](#).

Definition 1.2. $\mathbb{Z}_2^n := \mathbb{Z}_2 \times \dots \times \mathbb{Z}_2$. $\mathbb{Z}_2^n$ can be seen as the set of all functions from $\text{Fin } n$ to $\mathbb{Z}_2$. In this file, we consider $\mathbb{Z}_2^n$ as a [module](#) over $\mathbb{Z}_2$.

Definition 1.3. For $i \in \text{Fin } n$, let $e_i \in \mathbb{Z}_2^n$ be the canonical vector with 1 on the i -th position and zeros elsewhere

e (e)
$$e_i := (0, \dots, 0, 1, 0, \dots, 0),$$

e_i can be also seen as the [function defined on \$\text{Fin } n\$, supported at \$i\$, with value 1 there, and 0 elsewhere](#).

Note that $b + e_i$ is the code b with the i -th bit flipped, for every $b \in \mathbb{Z}_2^n$ and $i \in \text{Fin } n$.

Definition 1.4. We say that $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is a **strategy function**, if

SF (SF)
$$\forall b \in \mathbb{Z}_2^n, k \in \text{Fin } n, \exists i \in \text{Fin } n, f(b + e_i) = k.$$

We say that the puzzle **has a strategy** for n if

$$\exists f : \mathbb{Z}_2^n \rightarrow \text{Fin } n, \text{ (SF)}.$$

Definition 1.5. Let $\delta : \text{Fin } n \times \text{Fin } n \rightarrow \mathbb{N}$ be given by

d (δ)
$$\delta(i, j) = \text{if } i = j \text{ then } 1 \text{ else } 0.$$

2. MAIN RESULT

In this section, we characterize the values of n for which the puzzle has a strategy and we present a sketch of the proof. The details of the proof are given in the next sections and are adapted for formalization in Lean.

Theorem 2.1. *The puzzle has a strategy for $n \in \mathbb{N}^*$ if and only if n is a power of 2.*

Sketch of the proof.

Modus Ponens Let $k \in \text{Fin } n$. For every $b \in \mathbb{Z}_2^n$,

$\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n$ is surjective,

and thus also injective. So,

$$\begin{aligned} 2^n &= \sum_{b \in \mathbb{Z}_2^n} 1 = \sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = \sum_{i \in \text{Fin } n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b + e_i), k) \\ &= \sum_{i \in \text{Fin } n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = n \cdot \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k). \end{aligned}$$

Modus Ponens Reverse Let $n = 2^m$. Let $g : \text{Fin } (2^m) \rightarrow \mathbb{Z}_2^m$ be a bijection. Define

$$f(b) := g^{-1} \left(\sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right), \quad b \in \mathbb{Z}_2^n.$$

Let $b \in \mathbb{Z}_2^n$ and $k \in \text{Fin } n$.

$$\begin{aligned} f(b + e_i) = k &\Leftrightarrow \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + \sum_{j \in \text{Fin } 2^m} e_{ij} \cdot g(j) = g(k) \\ &\Leftrightarrow \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + g(i) = g(k) \Leftrightarrow i = g^{-1} \left(g(k) - \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right). \end{aligned}$$

3. LEAN TACTICS

- [apply](#)
- [by_contra](#)
- [constructor](#)
- [dsimp](#)
- [exact](#)
- [have](#)
- [intro](#)
- [let](#)
- [obtain](#)
- [rw](#)
- [simp \(only\)](#)
- [unfold](#)
- [use](#)

4. FUNCTION SHIFT IS INJECTIVE

`fun_shift_surj`

Lemma 4.1. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF) and $b \in \mathbb{Z}_2^n$, then $\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n$ is surjective.*

Proof. $\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n$ is [surjective](#), if

$$\forall k \in \text{Fin } n, \exists i \in \text{Fin } n, f(b + e_i) = k.$$

Taking our b in (SF), we get the desired conclusion. $\square$

`fun_shift_inj`

Lemma 4.2. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF) and $b \in \mathbb{Z}_2^n$, then $\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n$ is injective.*

Proof. Since $\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n$ is a [surjective](#) function, by Lemma 4.1, on a [finite](#) set, it is also [injective](#). $\square$

Mathlib hint: [Finite.injective_iff_surjective](#).

5. FUNCTION SHIFT SUM

`fun_shift_sum_one`

Lemma 5.1. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF), $b \in \mathbb{Z}_2^n$, and $k \in \text{Fin } n$, then*

$$\sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = 1.$$

Proof. Since f is (SF), we [obtain](#) $i_k \in \text{Fin } n$ such that $f(b + e_{i_k}) = k$.

We shall prove that [only the \$i_k\$ term in the sum is equal to 1, and the rest are 0](#).

Clearly, if $i = i_k$ in the above sum, then $\delta(f(b + e_i), k) = 1$, in view of (δ) .

Fix $i \in \text{Fin } n$ with $i \neq i_k$. Then, in view of (δ) , $\delta(f(b + e_i), k) = 0$, if $f(b + e_i) \neq k = f(b + e_{i_k})$. Assume that $f(b + e_i) = f(b + e_{i_k})$. Then, by Lemma 4.2, $i = i_k$, which [contradicts](#) the assumption that $i \neq i_k$. $\square$

Mathlib hints: `rw [Finset.sum_eq_single_of_mem i_k]`, `Finset.mem_univ`, `if_pos`, `if_neg`.

`fun_shift_sum_pow2`

Lemma 5.2. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF) and $k \in \text{Fin } n$, then*

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = 2^n.$$

Proof. By Lemma 5.1, for every $b \in \mathbb{Z}_2^n$, $\sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = 1$. So,

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = \sum_{b \in \mathbb{Z}_2^n} 1 = 2^n.$$

 $\square$

fun_shift_sum_rev

Lemma 5.3. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF) and $k, i \in \text{Fin } n$, then*

$$\sum_{b \in \mathbb{Z}_2^n} \delta(f(b + e_i), k) = \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k).$$

Proof. Since $\mathbb{Z}_2^n \ni b \mapsto b + e_i \in \mathbb{Z}_2^n$ is a [bijection](#), we the equality holds. $\square$

Mathlib hints: [Equiv.sum_comp](#), [Equiv.addRight](#).

6. MODUS PONENS

strategy_pow2

Lemma 6.1. *If $0 < n$ and $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF), then $\exists N \in \mathbb{N}, n \cdot N = 2^n$.*

Proof. Let $k := 0$ (which is in $\text{Fin } n$ since $0 < n$) in Lemma 5.2 to [have](#)

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = 2^n.$$

Since the [sums commute](#) and Lemma 5.3 holds,

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = \sum_{i \in \text{Fin } n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = n \cdot \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k).$$

We [use](#) $N := \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k)$. $\square$

mul_pow2

Lemma 6.2. *If $\exists N \in \mathbb{N}, n \cdot N = 2^n$, then $\exists m \in \mathbb{N}, n = 2^m$.*

Proof. Since $n \cdot N = 2^n$, we [have](#) n [divides](#) 2^n . Since [2 is prime](#), $\exists m \in \mathbb{N}, n = 2^m$. $\square$

Mathlib hint: [Nat.dvd_prime_pow](#).

puzzle_mp

Theorem 6.3. *If $0 < n$ and the puzzle has a strategy for n , then $\exists m \in \mathbb{N}, n = 2^m$.*

Proof. Since the puzzle has a strategy, $\exists f$ with (SF). By Lemma 6.1, $\exists N \in \mathbb{N}, n \cdot N = 2^n$. By Lemma 6.2, $\exists m \in \mathbb{N}, n = 2^m$. $\square$

7. MODUS PONENS REVERSE

bij_Fin

Definition 7.1. For $m \in \mathbb{N}$, let $g : \text{Fin } (2^m) \rightarrow \mathbb{Z}_2^m$ be a bijective function. g exists since $\text{Fin } (2^m)$ and $\mathbb{Z}_2^m$ are finite sets with the [same cardinality](#).

sum_basis

Lemma 7.2. If $i \in \text{Fin } n$ and $f : \text{Fin } n \rightarrow \mathbb{Z}_2^m$, then

$$\sum_{j \in \text{Fin } n} e_{ij} \cdot f(j) = f(i).$$

Proof. By (e), $e_{ij} = 1$ if $i = j$ and $e_{ij} = 0$ otherwise. $\sum_{j \in \text{Fin } n} e_{ij} \cdot f(j)$ simplifies to $f(i)$, since the single term that contributes is for $j = i$. $\square$

Mathlib hint: [rw\[Fintype.sum_single.smul\]](#).

puzzle_mpr

Theorem 7.3. If $n = 2^m$ for some m , then the puzzle has a strategy for n .

Proof. [Let](#) g be from Definition 7.1 and [use](#) $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ given by

$$f(b) := g^{-1} \left(\sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right), \quad b \in \mathbb{Z}_2^n.$$

Fix $b \in \mathbb{Z}_2^n$ and $k \in \text{Fin } n$. In view of (SF), we want to show $\exists i \in \text{Fin } n$, $f(b + e_i) = k$.

[Let's use](#) $i := g^{-1} \left(g(k) - \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right)$. By [distributivity of the scalar multiplication over addition](#) and [sum](#),

$$\sum_{j \in \text{Fin } 2^m} (b_j + e_{ij}) \cdot g(j) = \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + \sum_{j \in \text{Fin } 2^m} e_{ij} \cdot g(j).$$

By Lemma 7.2, $\sum_{j \in \text{Fin } 2^m} e_{ij} \cdot g(j) = g(i)$. By the expression of i ,

$$\begin{aligned} & g^{-1} \left(\sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + g(i) \right) \\ &= g^{-1} \left(\sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + g(k) - \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right) = g^{-1}(g(k)) = k. \end{aligned}$$

 $\square$

Mathlib hint: [g.symm](#).

8. FINISH

puzzle

Theorem 8.1. If $0 < n$, then the puzzle has a strategy for n if and only if $\exists m \in \mathbb{N}, n = 2^m$.

Proof. The forward direction is Theorem 6.3, and the reverse direction is Theorem 7.3. $\square$